Fraol Batole

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PhD Student — Software Engineering for AI | Coding Agents | Program Analysis

EDUCATION

Tulane University — Ph.D., Computer Science Aug 2024 – Dec 2026 (Expected)
New Orleans, Louisiana | GPA: 4.0/4.0 | Advisor: Prof. Hridesh Rajan

Iowa State University — M.Sc., Computer Science May 2024
Ames, Iowa | GPA: 3.62/4.0 | Advisor: Prof. Hridesh Rajan

Addis Ababa Science and Technology University — B.Sc., Software Engineering June 2018
Addis Ababa, Ethiopia | GPA: 3.75/4.0 | Graduated Magna Cum Laude | Founded Innovation Club

RESEARCH EXPERIENCE

Microsoft — Student Researcher (Contract) July 2026 – Oct 2026
Mentor: Ben Steenhoek, Manager: Gaurav Mittal, Yu Hu *Seattle, WA*

- Conducting applied research on coding agents with the GitHub Copilot team.

Laboratory for Software Engineering, Tulane University Aug 2024 – Present
Graduate Research Assistant *New Orleans, Louisiana*

- Built **RefineAct**, a runtime verification framework that derives formal specifications and enforces them during an LLM agent execution, enabling detection of failure modes invisible to static analysis or prompt engineering alone. ASE '26 (Distinguished Paper).
- **Agent-Oriented Design Issue Localization (LocalizeAgent):** Built a multi-agent framework integrating static analysis to produce abstraction-aware, LLM-friendly code summaries for automated design-issue localization. *ICSE '25.*
- **Typestate-Based Fault Localization:** Developed static analysis techniques to detect API usage violations in deep learning programs using typestate analysis. *TSE '26.*
- **LLM Agent Bug Study:** Led the first large-scale empirical study of LLM-based agent bugs, analyzing 1,016 bug reports and identifying tool integration (45%) and model integration (16%) as primary failure modes (under submission).
- **Move Method Refactoring (MM-Assist): JetBrains Collaboration:** Developed and released MM-Assist, an IntelliJ IDEA plugin for automated Move Method refactoring using LLMs, RAG, and hallucination filtering — achieving 73% accuracy and 4× improvement over prior tools (ICSME '25).

Polytechnique Montréal — Research Intern May 2025 – Aug 2025
Supervisor: Prof. Foutse Khomh *Montreal, Canada*

- Investigated the safety and robustness of LLM-based AI agents, focusing on novel detection and mitigation strategies for agent misbehavior. The work directly informs RefineAct.

TECHNICAL SKILLS

Programming Languages: Python, Java, Prolog
ML / LLM Frameworks: PyTorch, TensorFlow/Keras, Hugging Face Transformers
Agents & Tooling: LangChain, LangGraph, CrewAI, Model Context Protocol (MCP)
Program Analysis / SE: Static analysis, typestate analysis, runtime verification
Cloud / Infrastructure: Google Cloud Platform, Azure, National HPC clusters (ACCESS/XSEDE)

SELECTED PUBLICATIONS

Accepted

1. **Fraol Batole**, Ruchira Manke, Tien N. Nguyen, and Hridesh Rajan. “SpecTrace: Automatically Auditing the MCP Tool Servers Behind AI Agents for Protocol Compliance.” ICSE’27
2. **Fraol Batole**, Foutse Khomh, and Hridesh Rajan. "Automatic Runtime Verification of LLM Agent Actions." ASE’26 (Won Distinguished Paper Award)
3. **Fraol Batole**, Ruchira Manke, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "Typestate-Based Fault Localization of API Usage Violations in Deep Learning Programs." *TSE '26*.
4. **Fraol Batole**, David O'Brien, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "An LLM-Based Agent-Oriented Approach for Automated Code Design Issue Localization." *ICSE '25*.

5. **Fraol Batole**, Ruchira Manke, David O'Brien, Steven Vo, Tien N. Nguyen, and Hridesh Rajan. "An Empirical Study of Bugs in LLM-based Agent Frameworks" Major Revision at EMSE.
6. Abhiram Bellur, Mohammed Raihan Ullah, **Fraol Batole** et al. "Multi-Agent Coordinated Rename Refactoring." FSE'26 (Tool Demo Track)
7. Abhiram Bellur, **Fraol Batole,** et al. "LLM, IDE and Semantic Embedding to Assist Move Method Refactoring." *ICSME '25*.
8. Sayem Imtiaz, Astha Singh, **Fraol Batole**, and Hridesh Rajan. "IRepair: An Intent-Aware Approach to Repair Data-Driven Errors in LLMs." *FSE '25*.
9. Sayem Imtiaz, **Fraol Batole,** et al. "Decomposing a recurrent neural network into modules for enabling reusability and replacement" *ICSE '23*.

Under Submission

10. **Fraol Batole,** Abdulmajid Awol Seid, Tien N. Nguyen, Foutse Khom, and Hridesh Rajan. " **Detecting Compositional Harm in LLM-based Agent Actions"**
11. **Fraol Batole**, Smit Patel, Aashish Yadavally, Tien N. Nguyen, and Hridesh Rajan. "CAPS: Code Abstraction Pre-training Strategy for Large Language Models."
12. Samuel W. Flint, Robert Dyer, **Fraol Batole**, and David O'Brien. "Do LLMs Generate Code with Type Annotations?"
13. David O'Brien, **Fraol Batole** et al. "Issue-Based Programming Language Evolution."

ACADEMIC SERVICE & AWARDS

Distinguished Reviewer Award (Junior PC) — Mining Software Repository (MSR) '26 April 2026

Reviewer / Program Committee:

- Reviewer — Transactions on Software Engineering (TSE '26)
- Junior Program Committee (Technical Track) — Mining Software Repositories (MSR '26)
- Reviewer — Transactions on Software Engineering and Methodology (TOSEM '26)
- Reviewer — Automated Software Engineering (ASE) Journal '25
- Reviewer — Deep Learning for Code Workshop @ NeurIPS '25

ADDITIONAL EXPERIENCE

Iowa State University — Graduate Teaching Assistant Jan 2021 – Aug 2024

CS Department *Ames, Iowa*

- Taught programming principles, debugging, and test-case development to 80+ students. Adapted pedagogy for diverse learning styles.

MED Innovation — Software Engineer Aug 2019 – Nov 2019

Addis Ababa, Ethiopia

- Integrated a CRM system for a cross-functional team using Spring Boot and Vaadin.

HONORS & GRANTS

- **NSF-Funded Advanced Computing Credits:** 1.5 million credits awarded (Feb 2024, Jun 2025).
- **Google Cloud Research Credits:** $12,000 awarded (Jun 2023).
- **Google Cloud Research Innovator '23:** Competitively selected for Google's research program (Mar 2023).
- **Magna Cum Laude, B.Sc. Software Engineering**: Addis Ababa Science and Technology University.